

Post-Quantum Identity-Based Linkable Ring Signcryption for Edge IoT Devices

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Abstract—With the rapid expansion of consumer electronics and the Internet of Things (IoT) ecosystem, ensuring secure communication in resource-constrained edge devices has become a critical challenge, especially under the threat of quantum attacks. Although lattice-based cryptographic schemes provide post-quantum security, they often suffer from large key storage and high computational overhead, limiting their applicability in edge environments. To address these challenges, we propose an identity-based linkable ring signcryption scheme constructed on the post quantum Lattice cryptography, tailored for edge-enabled IoT systems. The scheme employs a trapdoor generation algorithm to establish the master keys, utilizes compact Gaussian sampling to derive private keys for ring members, and integrates rejection sampling to generate user signatures with strong anonymity. Furthermore, a key encapsulation mechanism ensures identity protection during encryption. Security analysis under the Random Oracle Model (ROM) demonstrates the scheme's resistance to forgery and its linkability property. Performance evaluation shows that the proposed approach achieves smaller public keys, reduced encryption/decryption costs, and enhanced efficiency, making it well-suited for lightweight IoT and consumer electronics applications in the post-quantum era.

Index Terms—Post-quantum cryptography, edge IoT devices, linkable ring signcryption, identity-based encryption, rejection sampling, consumer electronics security.

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I. INTRODUCTION

SIGNCRYPTION schemes provide the combined functionality of public-key encryption and digital signatures in a single logical step. Compared with the conventional “sign-then-encrypt” approach, signcryption significantly reduces both computational and communication overhead, making it an essential technique for lightweight identity authentication. These properties make signcryption particularly relevant in domains such as electronic cash payments and secure authentication [1], [2].

Ring signcryption schemes [3], which merge the concepts of ring signatures and signcryption, further enhance security by ensuring confidentiality, authenticity, and anonymity of the sender. In edge-enabled IoT contexts, identity-based cryptography (IBC) and linkable ring signcryption are integrated for a number of important reasons. First, memory, processing, and energy are severely limited in the operation of IoT and edge devices. Because of the creation, distribution, and verification of certificates, traditional certificate-based public-key systems have significant overhead. By doing away with certificates and enabling devices to utilise their identity as a public key, identity-based encryption streamlines key management and boosts scalability in extensive IoT installations. Second, security primitives that simultaneously guarantee anonymity, secrecy, and authenticity are necessary for edge IoT systems. Linkable ring signcryption is appropriate for latency-sensitive environments since it offers these characteristics in a single, lightweight operation. To guarantee functionality, scalability, and resilience in limited contexts, IoT performance validation is carried out utilizing a combination of simulation and experimental (testbed) approaches. In order to replicate real-world settings, these techniques require meticulous parameter, message length, and hardware configuration. utilized as “ground truth” for simulator calibration to confirm functionality, compatibility, and real-world performance with actual hardware. During this stage, pilot projects are frequently used to assess different hardware and physical characteristics. Castiglione et al. [4] introduced a privacy-preserving method that combines ring signcryption with identity-based cryptography, assigning tracking tags to identify misbehaving nodes. In parallel, Choudhury et al. [5] proposed a certificateless ring signcryption (CLGRSC) scheme based on bilinear pairings, while Ebrahimi et al. [6] developed an elliptic curve-based certificateless approach supporting conditional privacy protection. Commey et al. [7] demonstrated weaknesses in the

scheme of [8] and proposed an improved conditional privacy-preserving (CPP) variant for VANETs. Similarly, Yacoub Hanna et al. [9] designed an anonymous linkable ring sign-encryption (ALRS) scheme for location-based services (LBS) in vehicular networks to address privacy concerns of vehicle users and service providers [3], [10].

With the advent of quantum computing, the security guarantees of conventional ring sign-encryption schemes are at risk. The vulnerability of conventional number-theoretic systems is effectively addressed by post-quantum lattice-based cryptography, which relies on difficult mathematical problems that are unsolvable even for quantum computers. Discrete logarithms and integer factorisation are issues that Shor's method can effectively handle on a suitably powerful quantum computer. Because of this, in the post-quantum age, standard public-key systems are inherently insecure. On the other hand, high-dimensional lattice calculations are the foundation of lattice-based cryptography, such as Learning with Errors (LWE), Ring-LWE, and NTRU-SIS. Since there are no known quantum algorithms that can solve these issues in polynomial time, it is thought that they are resistant to both classical and quantum attacks. Lattice-based cryptography has emerged as a leading candidate in the post-quantum era, offering resistance to quantum attacks, high computational efficiency, and flexible construction options. Since lattice-based systems rely primarily on matrix–vector and polynomial operations, they exhibit better asymptotic efficiency and parallelism compared to number-theoretic cryptography [11]. In recent years, several lattice-based sign-encryption schemes [12], [13] have been developed. For example, Sedghighadikolaei et al. [14] introduced a lattice-based blind sign-encryption scheme, while Malina et al. [15] presented a certificateless, quantum-resistant ring sign-encryption scheme grounded in SIS and LWE assumptions. The cryptographic task shifts from hard algebraic inversion to sampling from high-dimensional discrete Gaussians and a few linear algebra operations using trapdoor-based lattice structures. The expense of creating a high-quality trapdoor basis (offline), the cost of discrete-Gaussian preimage sampling per signature (online), and related large-dimension linear algebra (matrix–vector multiplies, in ring module variations) are the practical bottlenecks. Using fast convolution-based sampling, converting to structured RLWE module lattices and gadget trapdoors, improving trapdoor quality (so the sampler requires smaller Gaussian widths), and carefully engineering the sampler convolution, precomputation, vectorisation. The smoothing-parameter lower bound must be respected by all optimisations in order to prevent entropy leakage and security breaches. Irshad et al. [16] further extended this line of work by designing a consortium blockchain-oriented lattice-based ring sign-encryption scheme, where smart contracts govern the sign-encryption and unsign-encryption processes.

Despite their advantages, lattice-based signature and sign-encryption schemes often suffer from high computational and communication costs. The post quantum Lattice cryptography, defined over polynomial rings, provides a more efficient alternative by enabling shorter key lengths and compact signatures, thereby making it well-suited for resource-constrained environments such as IoT-enabled consumer

electronics [17]. The security reduction that links the resilience of a cryptographic method is a mathematical demonstration that it is at least as hard to break the scheme's security as it is to solve some known, underlying hard problem. In essence, this demonstration converts the challenge of compromising the scheme (such as forging a signature or decrypting a message without a key) into an example of a problem that is generally thought to be difficult (such as factoring huge numbers, the Discrete Logarithm issue, or lattice problems). To address efficiency limitations, many works have also explored linkability [18], [19], [20], which allows verification of whether two or more signatures originate from the same signer while preserving anonymity and unforgeability. This additional property has been widely adopted in privacy-sensitive applications. Mathematical problems for which there are no known efficient classical or quantum methods, post-quantum lattice-based cryptography provides robust defence against both classical and quantum adversaries. The Learning with Errors (LWE) problem, one of the most commonly accepted hardness assumptions in post-quantum cryptography, is essential to its security. Recovering a secret vector from noisy linear equations is the LWE issue. The purposefully added noise renders the problem computationally unsolvable even with complete understanding of the equations' structure. Since LWE is not susceptible to Shor's technique, quantum computers do not provide a substantial advantage in solving it, in contrast to number-theoretic issues employed in RSA or ECC.

Several NTRU-based approaches have been proposed to enhance linkability and efficiency. Sezer et al. [21] introduced Raptor, a linkable ring signature scheme using chameleon hash functions. By enabling domain separation and layered binding of messages, identities, and public parameters, multiple hash functions in cryptographic methods improve security. By calculating unique hashes across distinct concatenations or prefixes, they bind these elements. Concatenated or parallel outputs retain resistance equal to the most secure function, minimising single-hash vulnerabilities like MD5 collisions, while many hashes strengthen collision resistance by forcing adversaries to break the strongest function. Diversity, other hashes maintain integrity in the event of a collision assault, lowering the effective attack surface. Through the use of a secret “trapdoor,” trapdoor sampling algorithms in key generation, particularly for lattice-based cryptography, enable the efficient generation of short, random-looking keys (vectors) required for schemes like digital signatures and encryption. This ensures strong unpredictability while drastically reducing the complex sampling of truly random vectors (like Gaussians). The trapdoor allows easy inversion (sampling) for the holder but remains computationally difficult for others, balancing performance and security by simplifying complex lattice operations. Wang et al. [22] optimized signing efficiency with rejection sampling and trapdoor generation on post quantum Lattice cryptography. Chawla et al. [23] further addressed signature length and generation efficiency issues by designing an NTRU-based linkable ring signature (LRS) scheme with tag-based linkability, improving upon Torres's scheme [24]. Similarly, Cao et al. [25] proposed a

hybrid approach leveraging Verifiable Pseudorandom Functions (VPFs) and quantum-secure NTRU-based linkable ring signatures to strengthen robustness.

Motivated by these advancements, this work proposes an *identity-based linkable ring signcryption scheme over post quantum Lattice cryptography* to enhance computational efficiency while maintaining strong post-quantum security guarantees. The main contributions are as follows:

- 1) By integrating post quantum Lattice cryptography with identity-based linkable ring signcryption, we formally prove under the Random Oracle Model (ROM) that the scheme achieves confidentiality, unforgeability, unconditional anonymity, and linkability. Confidentiality, authenticity, and anonymity formal security models usually use game-based definitions in which a challenger engages with a probabilistic polynomial-time (PPT) opponent. In order to try to breach security properties, the adversary takes control of the network, intercepts and alters communications, and queries oracles (such as encryption, signature, or session disclosure). Computational time, storage overhead, communication cost, and key size are evaluation criteria used to determine how well a given scheme performs in Internet of Things (IoT) application. These factors are essential for figuring out whether the scheme is appropriate for IoT contexts and devices with limited resources. Effective communication saves bandwidth, which is sometimes scarce in extensive IoT deployments, and lessens network congestion. Additionally, it reduces data transmission time, which has a direct effect on power usage and latency. If the enemy has a very small chance of prevailing, security is maintained. Formalise confidentiality, which guarantees that an attacker cannot decipher encrypted secrets.
- 2) We utilize the Compact Gaussian Sampler (CGS) algorithm [26] along with rejection sampling techniques [27] to efficiently generate user private keys and signatures, thereby improving key generation performance.
- 3) We conduct detailed performance analysis in terms of time complexity and storage overhead. Results demonstrate that the proposed scheme achieves superior efficiency compared to existing lattice-based ring signcryption solutions, making it particularly suitable for IoT consumer electronics in the post-quantum era.

The remainder of this paper is organized as follows. Section II introduces the preliminaries, including symbol descriptions, lattice definitions, and hardness assumptions that form the basis of the proposed construction. Section III defines the security model and outlines the identity-based linkable ring signcryption scheme on the NTRU lattice. Current ring signature and signcryption systems have a number of significant drawbacks when trying to accomplish both accountability and privacy at the same time. By hiding the signer within a group, traditional ring signatures offer excellent anonymity. However, this very feature makes it challenging to identify criminal activity or recurrent misuse by the same person. Because of this, these programs frequently sacrifice accountability in order to protect privacy. Similar to this, many signcryption techniques do not include means for regulated traceability and

TABLE I
LIST OF SYMBOLS

Symbol	Meaning
$R = \mathbb{A}[x]/\langle x^n + 1 \rangle$	Polynomial ring
R^m	m -dimensional vector space
$R_q = \mathbb{A}_q[x]/\langle x^n + 1 \rangle$	Polynomial ring modulo q
B	Matrix
x	Column vector
$x \leftarrow E$	Vector x chosen from probability distribution D
$\ x\ $	Euclidean norm of vector x
$\mathbb{R}$	Set of real numbers
$\mathbb{A}$	Set of integers
$\mathbb{A}_q^n$	n -dimensional vector space modulo q
$\mathbb{A}_q^{n \times m}$	$n \times m$ matrix space modulo q
$f \cdot g$	Polynomial multiplication
$\text{negl}(n)$	Negligible function in n

instead concentrate on merging confidentiality and authenticity. As a result, they are not able to recognise or connect messages that come from the same signer, which limits their applicability in situations where both misbehaviour detection and privacy protection are necessary. Section IV presents the proposed post-quantum lattice-based scheme in detail, covering the system design, initialization, key generation, signcryption, and unsigncryption processes. Section V analyzes the correctness of the scheme, while Section VI provides formal security proofs with respect to confidentiality, unforgeability, anonymity, and linkability. Sections VII and VIII discuss the anonymity and linkability properties in more depth and evaluate the performance of the proposed approach compared to existing schemes. Section IX highlights the results and discusses their implications for IoT consumer electronics. Finally, Section X concludes the paper and outlines directions for future research.

II. PRELIMINARIES

A. Symbol Description

Table I describes the symbols used in this paper.

III. POST-QUANTUM LATTICE CRYPTOGRAPHY

This section introduces the mathematical framework and algorithmic foundations of post-quantum lattice cryptography, which forms the basis of our proposed scheme. Instead of traditional definitions, we present the concepts as computational algorithms to highlight their practical cryptographic relevance. Traditional public-key infrastructure (PKI)-based cryptosystems mostly rely on certificate production, distribution, validation, and revocation, they are frequently inappropriate for IoT devices with limited resources. Low-power IoT devices, like wearables, sensors, and embedded edge nodes, are usually unable to handle the substantial computational, communication, and storage overhead that these operations impose. Compared to Certificate-Based Encryption (CBE) and Attribute-Based Encryption (ABE), Identity-Based Encryption (IBE) significantly improves deployment simplicity for Internet of Things applications by doing away with complicated Public Key Infrastructure (PKI) management. However, this is compromised by difficulties managing revocations effectively and an overall greater computational load. A web transfer

Algorithm 1 Lattice Construction**Require:** n linearly independent vectors $b_1, b_2, \dots, b_n \in S^m$ **Ensure:** Lattice Λ

1: Compute lattice as:

$$\Lambda \leftarrow \left\{ \sum_{i=1}^n x_i b_i \mid x_i \in \mathbb{Z} \right\}$$

2: **return** Λ **Algorithm 2** NTRU Lattice Generation**Require:** Prime $r > 2$, power-of-two n , polynomials $f, h \in S_q$ **Ensure:** NTRU lattice $\Lambda_{g,r}$ and its basis matrix1: Compute $g \leftarrow h \cdot f^{-1} \bmod q$

2: Construct lattice:

$$\Lambda_{g,r} \leftarrow \{(v, w) \in S^2 \mid v + w \cdot g \equiv 0 \pmod{r}\}$$

3: Define basis matrix:

$$\Lambda_{g,r} = \begin{pmatrix} -M_n(g) & I_n \\ rI_n & O_n \end{pmatrix}$$

4: **return** $\Lambda_{g,r}$ **Algorithm 3** Discrete Gaussian Sampling (DGS)**Require:** Lattice Λ , Gaussian width σ , center vector $b \in S^m$ **Ensure:** Sample $x \in \Lambda$

1: Compute probability distribution:

$$P(x) = \frac{\rho_{b,\sigma}^m(x)}{\sum_{y \in \Lambda} \rho_{b,\sigma}^m(y)}, \quad \text{where}$$

$$\rho_{b,\sigma}^m(x) = \exp(-\pi \|x - b\|^2 / \sigma^2)$$

2: Sample $x \sim P(x)$ 3: **return** x

protocol designed for usage in limited devices and networks is called limited Application Protocol (CoAP). This makes it appropriate for Internet of Things devices that run on sporadic networks or have limited battery life. Additionally, managing digital certificates necessitates regular updates, constant communication to certificate authorities, and extra memory to hold certificate chains—all of which clash with IoT hardware's limited processing power and short battery life. The section covers lattice construction, NTRU lattice generation, discrete Gaussian sampling, hardness assumptions (RLWE, DRLWE, and NTRU-GSIS), and key auxiliary algorithms such as trapdoor generation, compact Gaussian sampling, and rejection sampling.

A lattice Λ is the discrete additive subgroup generated by integer linear combinations of its basis vectors. Here, n and m denote the rank and dimension, respectively.

The NTRU lattice achieves compact storage due to polynomial representation and is widely used in constructing trapdoors for cryptographic operations.

Computation, storage, and decision-making are increasingly carried out at or close to the data source in edge-driven architectures, which dramatically lowers latency and bandwidth consumption while also imposing stringent limitations

Algorithm 4 Ring Learning With Errors (RLWE)**Require:** Ring size n , modulus q , error distribution χ **Ensure:** Challenge pair (a, b) 1: Sample $a \leftarrow S_q$ uniformly2: Sample error $e \leftarrow \chi$ 3: Compute $b \leftarrow a \cdot x + e$, where $x \in S_q$ is hidden4: **return** (a, b) **Algorithm 5** Decision-RLWE (DRLWE)**Require:** Ring size n , modulus $q = 2^k$, error distribution χ **Ensure:** Challenge pair (c, b) 1: Choose $a, x \leftarrow S_q$ uniformly2: Sample error $e \leftarrow \chi$ 3: Compute $b \leftarrow c \cdot x + e$ 4: With probability $1/2$, output (c, b) ; otherwise output a uniform sample from $S \times S_q$

on processing power, memory, and energy. The design philosophy of contemporary cryptographic primitives is also strongly impacted by this decentralisation: security mechanisms must be sufficiently light to function effectively on edge nodes with low resources while maintaining robust assurances of secrecy, integrity, and authentication. Low-complexity algorithmic structures, hardware-friendly lattice- or hash-based constructions, and communication-efficient protocols that preserve strong security without overtaxing edge devices or jeopardising system-wide confidence are being given priority in current research. The trade-offs between unlikability and accountability in identity-based cryptographic systems centre on striking a balance between a user's right to privacy and the requirement for system monitoring and the capacity to link activities to particular people when needed. Improved privacy for users. Users can interact with the system without their identity being readily revealed or their behaviours being connected. Applications handling sensitive data, whistleblowing, or safeguarding user browsing patterns must do this. Because anonymity makes it harder to identify undesirable actors, it may be possible to engage in illegal transactions, malware distribution, or spamming without worrying about the consequences. DGS is critical for generating short vectors in lattices, ensuring secure key material in cryptographic systems.

A. Hard Problems in Lattice Cryptography

The security of lattice-based schemes relies on problems that remain computationally infeasible even for quantum adversaries. These are described algorithmically as follows.

The RLWE problem asks whether (a, b) can be distinguished from a uniform random pair in $S_q \times S_q$.

The DRLWE problem requires distinguishing whether a given pair comes from RLWE distribution or uniform distribution.

The hardness assumption states that no polynomial-time adversary can solve NTRU-GSIS without trapdoor knowledge.

Algorithm 6 NTRU Small Integer Solution (NTRU-GSIS)**Require:** Polynomial $g = h \cdot f^{-1} \bmod q \in S_q$, bound $\beta > 0$ **Ensure:** Non-zero small polynomials $(v, w) \in S_q^2$ 1: Find $(v, w) \neq (0, 0)$ such that:

$$v + w \cdot g \equiv 0 \pmod{q}, \quad \|v\|_2, \|w\|_2 \leq \beta$$

2: **return** (v, w) **Algorithm 7** Trapdoor Generation (TrapGenNTRU)**Require:** Prime $q \equiv 1 \pmod{2n}$, $n = 2^k$, Gaussian parameter $\sigma_f = 1.17q/2n$ **Ensure:** Polynomial g and short basis $B_{f,h}$ 1: Select $f, h, F, H \in S$ such that $f \cdot H - h \cdot F = q$ 2: Compute $g \leftarrow h \cdot f^{-1} \bmod q$ 3: **return** $(g, B_{f,h})$ as the trapdoor basis**Algorithm 8** Compact Gaussian Sampler (CGS)**Require:** Basis $B = \{b_1, \dots, b_n\}$, Gaussian parameter σ , center $d \in A^m$ **Ensure:** Sample z 1: Sample $z \sim E_{\Lambda(B), \sigma, d}$ 2: **return** z *B. Auxiliary Algorithms for Lattice-Based Cryptography*

The following algorithms enable efficient key generation, signature sampling, and security guarantees in our scheme.

CGS ensures efficient and statistically close approximation to ideal discrete Gaussian distributions, enabling lightweight implementations for IoT.

Rejection sampling guarantees that the distribution of signing keys is statistically independent of the signatures, which is essential for preserving anonymity and preventing information leakage.

Together, these algorithms form the backbone of the proposed lattice-based identity-based linkable ring signcryption scheme, providing both efficiency and post-quantum security guarantees.

C. Identity-Based Linkable Ring Signcryption Scheme on NTRU Lattice

NTRU-IBRSC consists of five PPT algorithms:

- 1) **Setup**($1^\lambda, 1^N$): Input the security parameter λ and the number of ring members N , output the public parameters params and the master secret key MSK.
- 2) **KeyGen**($\text{params}, ID_i, \text{MSK}$): Input the public parameters params , the user identity ID_i , and the system master secret key MSK, output the user's public/private key pair (pk_i, sk_i) .
- 3) **RingSigncryption**($\text{params}, V, m, sk_x, pk_x$): Input the public parameters params , the set of ring user identities $V = (ID_1, ID_2, \dots, ID_N)$, the message $m \in \{0, 1\}^\tau$, and the private key $sk_x = (s_1, s_2, s'_1, s'_2)$ and public key pk_x of the actual signer ID_x , output the ciphertext $D = (v, w, d, \text{Sig}(m))$, which includes the ring signature $\text{Sig}(m)$ of user ID_x .

Algorithm 9 Rejection Sampling for Signature Generation**Require:** Vector $w \in A^m$, Gaussian parameter σ , constant $M = O(1)$ **Ensure:** Sample z independent of w 1: **repeat**2: Sample $z \sim E_\sigma^m$ 3: Accept z with probability

$$\min\left(\frac{E_\sigma^m(z)}{M \cdot E_{w, \sigma}^m(z)}, 1\right)$$

4: **until** accepted5: **return** z

- 4) **Unsigncryption**($\text{params}, V, m, s_2, D, \text{event}$): Input the public parameters params , the set of ring user identities $V = (ID_1, ID_2, \dots, ID_N)$, the message $m \in \{0, 1\}^\tau$, the receiver's partial private key s_2 , the ciphertext $E = (v, w, d, \text{Sig}(m))$, and the event identifier event . If successful, output the message m ; otherwise, output $\perp$.
- 5) **Link**($\text{Sig}(m_1), \text{Sig}(m_2)$): Input two signatures $\text{Sig}(m_1)$ and $\text{Sig}(m_2)$ containing link tags J_1 and J_2 . The algorithm first verifies the validity of both signatures, then checks whether $J_1 = J_2$. If both conditions are satisfied, output "Linked", indicating that both signatures are generated by the same signer; otherwise, output "Unlink", indicating that the two signatures are generated by different signers.

IV. SECURITY MODEL

The suggested integration is a significant and timely innovation that integrates three complementary concepts ring linkability (selective anonymity with accountable linking), post-quantum hardness (resistance to quantum adversaries), and identity-based key management (reducing PKI overhead) into a single primitive designed for mobile, constrained edge-IoT networks. When combined, these components tackle a number of actual issues with edge deployments, including the difficulty of key distribution, upcoming quantum threats, and the competing demands of privacy and auditability in group communication. The proposed NTRU lattice-based identity-based linkable ring signcryption scheme (NTRU-IBLRSCS) is designed to secure lightweight consumer IoT devices operating in post-quantum environments. By leveraging hardness assumptions such as DRLWE and NTRU-SIS, together with efficient algorithms like trapdoor generation, compact Gaussian sampling (CGS), and rejection sampling, the scheme guarantees *confidentiality*, *unforgeability*, *unconditional anonymity*, and *linkability*. These properties are formally modeled through a sequence of interactive games between an adversary δ and a challenger γ , simulating real-world attack scenarios in resource-constrained IoT ecosystems [28], [29]. The following oracles are used in defining the security model:

- 1) **Registration Oracle (RO)**: On input of a user identity ID_i , the oracle returns the corresponding public key pk_i .
- 2) **Corruption Oracle (CO)**: On input of a user identity ID_i , the oracle outputs the corresponding private key sk_i , simulating device key compromise in IoT systems.

- 3) **Signcryption Oracle (SO):** Given a tuple $(V, m \in \{0, 1\}^*, sk_i)$, where $V = (ID_1, ID_2, \dots, ID_N)$ is the ring, m is a message, and sk_i is the private key of $ID_i \in U$, the oracle generates a valid signcryption ciphertext using `RingSigncryption`.

A. Confidentiality

Confidentiality is modeled through the anonymity of ciphertext indistinguishability under chosen-identity and chosen-plaintext attacks (ANON-sID-CPA). The master secret key (MSK) from a high-entropy space, which is usually a random scalar in pairing-based systems or a big integer or vector selected uniformly at random in lattice-based systems. Since only the PKG is aware of this MSK, it becomes the central authority that manages the extraction of all user keys, making it impossible for any unauthorised party to create private keys or forge identities. The master public key (MPK) is then calculated by the PKG using the system's hard mathematical function, such as multiplying the MSK by a generator in bilinear-pairing groups or publishing a set of public matrices formed from the MSK in lattice-based structures. Anyone can encrypt or validate signatures using this MPK, which is broadcast throughout the system, without the requirement for conventional certificates. The game between δ and γ proceeds as follows:

- 1) **Initialization:** γ runs `Setup`($1^\lambda, 1^N$) (Algorithm 7) and publishes the master public key h and parameters $params$, keeping the trapdoor-based master secret key private.
- 2) **Query 1:** δ submits ID_i . γ computes sk_i using `KeyGen`($params, ID_i, MSK$) (Algorithm 8).
- 3) **Challenge:** δ submits (ID_0^*, ID_1^*, m) . γ selects a random bit $b^* \in \{0, 1\}$ and returns

$$C^* \leftarrow \text{RingSigncryption}(params, ID_{b^*}^*, m, sk_x, pk_x). \quad (1)$$

- 4) **Query 2:** For all $ID_i \neq \{ID_0^*, ID_1^*\}$, δ may query `KeyGen` adaptively.
- 5) **Guess:** δ outputs b . Success occurs if $b = b^*$.

The advantage is defined as:

$$\text{Adv}_A^{\text{ANON-sID-CPA}} = \Pr[b^* = b] - \frac{1}{2}. \quad (2)$$

If this advantage is negligible, the scheme ensures confidentiality even in post-quantum IoT settings.

B. Unforgeability

Unforgeability ensures that no adversary can generate a valid signcryption without access to the legitimate keys. The adversarial game is:

- 1) γ runs `Setup`($1^\lambda, 1^N$) and shares $h, params$.
- 2) δ adaptively queries RO, CO, and SO.
- 3) δ outputs $(m^*, V^*, \text{Sig}(m^*))$. The forgery is valid if:
 - a) $\text{Sig}(m^*)$ verifies correctly,
 - b) all keys in U^* were obtained via RO,
 - c) no CO query was made for V^* ,
 - d) $\text{Sig}(m^*)$ was not generated by SO.

This property directly relies on the hardness of the NTRU-SIS problem, ensuring robustness against post-quantum adversaries.

C. Unconditional Anonymity

The scheme preserves anonymity such that even a computationally unbounded adversary cannot distinguish the true signer among ring members with probability exceeding $1/2$. The anonymity game is defined as:

- 1) γ initializes the system with `Setup`.
- 2) δ queries RO, CO, and SO.
- 3) δ selects two identities ID_0, ID_1 and a message m . γ computes:

$$\text{Sig}(m^*) \leftarrow \text{RingSigncryption}(params, V, m, sk_x, pk_x). \quad (3)$$

- 4) δ guesses the signer. Advantage is defined as:

$$\text{Adv}_{\text{anon}}^\delta = \Pr[b^* = b] - \frac{1}{2}. \quad (4)$$

D. Linkability

Linkability ensures that two signatures generated by the same IoT device can be efficiently linked through their tag J . The adversarial game is:

- 1) γ initializes the scheme.
- 2) δ performs oracle queries.
- 3) δ outputs two valid signatures $(m_1^*, V_1^*, \text{Sig}_1)$ and $(m_2^*, V_2^*, \text{Sig}_2)$.
- 4) δ succeeds if:
 - a) both signatures are valid,
 - b) $\text{Link}(\text{Sig}_1, \text{Sig}_2) = \text{"Unlinked"}$,
 - c) neither was produced by SO,
 - d) CO was queried at most once.

The adversary's advantage is:

$$\text{Adv}_{\text{link}}^\delta = \Pr[\delta \text{ wins the game}]. \quad (5)$$

If $\text{Adv}_{\text{link}}^\delta < \text{negl}(n)$, the scheme is linkable, enabling efficient detection of repeated signers in IoT consumer networks.

V. POST QUANTUM LATTICE CRYPTOGRAPHY-BASED IDENTITY-BASED LINKABLE RING SIGNCRYPTION SCHEME

A. System Design

In order to further improve the computational efficiency of lattice-based ring signcryption schemes, the identity-based ring signcryption scheme on lattices is extended to the post quantum Lattice cryptography, while introducing the linkable property. In order to balance security and efficiency, the Setup phase of lattice-based cryptography entails choosing the lattice dimension, modulus, and error boundaries. This balance permits the cryptographic technique to be useful for practical applications while guaranteeing that the underlying hard issues (such as the Learning with Errors (LWE) problem) are computationally impractical for attackers, even those utilizing quantum computers. The Setup phase of lattice-based cryptography entails choosing system parameters, such as the

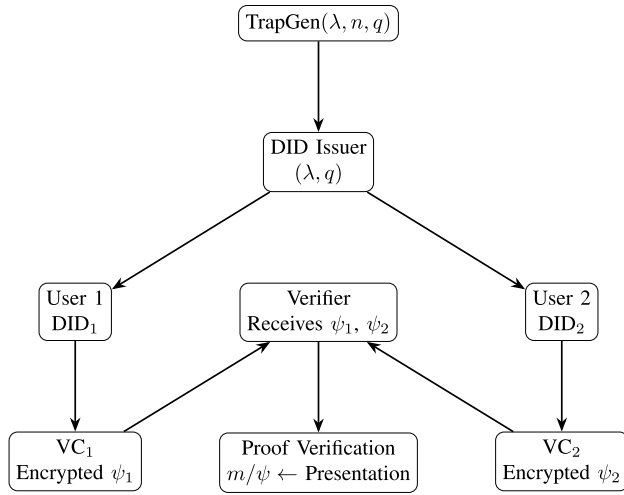


Fig. 1. System model diagram (alternative layout).

lattice dimension, modulus, and error boundaries, in order to balance efficiency and security. In practical applications, this equilibrium guarantees the fundamental challenges. The system model is shown in Figure 1.

In the setup phase, the system's master key is generated using the trapdoor generation technique on the post quantum Lattice cryptography. Since the worldwide system values are obtained from fixed-dimension lattice bases and modulus parameters that are independent of the number of registered users, the scheme maintains constant-size public parameters and ciphertexts. By mapping user identities to points in a structured algebraic space (for example, through hash-to-lattice approaches), identity-based and lattice-based constructs of this type enable the system to implicitly incorporate identification information instead of increasing public parameters per user. Consequently, as the population grows, neither the ciphertext format nor the public key material do. In the key generation phase, the compact Gaussian sampling (CGS) algorithm is used to generate private keys, thereby accelerating user key extraction. A trusted organization known as the Private Key Generator (PKG) handles the Key Extraction (or Key Generation) stage of Identity-Based Encryption (IBE) systems. This phase's basic idea is to utilize a user's unique identity string (such as their email address) as their public key and then use the PKG's master secret key to determine the matching private key. The sampling processes focus on leveraging the PKG's master secret to deterministically construct a unique, mathematically connected private key for a given identity rather than collecting biometric data or physical signals. Provide confidentiality and authenticity at a lower computational cost, classical signcryption effectively combines encryption and digital signatures in a single process. However, since the sender's identity is always openly linked to the signcrypted message, it does not offer anonymity. By enabling a signer to blend in with a group of potential signers, ring signatures solve the anonymity problem. Although the verifier cannot identify the signer, they can verify that a member of the ring provided the signature. Strong privacy is guaranteed, but accountability is eliminated because it is impossible to identify deliberate or

persistent misconduct by the same organisation. By allowing regulated traceability while maintaining anonymity, linkable ring signcryption offers a compromise. Verifiers can determine whether several communications come from the same anonymous sender by using a cryptographic link tag. In the signcryption phase, rejection sampling is employed to generate signatures with a certain probability, encryption is performed using the key encapsulation mechanism [25], and a hash function is applied as a one-time pad for the message.

The detailed description of the system model in Figure 1 is as follows:

B. Setup($1^\lambda, 1^N$) System Initialization

Input security parameter λ and number of ring members N . The PKG performs as follows:

- 1) Select an integer $n = 2^k$ and τ , where $k > 0$, τ denotes the length of message m . Then select a prime satisfying $q \equiv 1 \pmod{2n}$, two parameters $\sigma_f = 1.17q/2n$ and Gaussian parameter

$$\sigma = \frac{117}{200\pi} \cdot q \log \exp \left(2 + \frac{2}{\eta} \right), \quad (6)$$

where $\eta = (2 - (\lambda + 1)/n)$.

- 2) Run the trapdoor generation algorithm $TrapGen_{NTRU}(q, n, \sigma_f)$ to generate a basis on lattice $\Lambda_{g,q}$, outputting $\{g \in R_q, B \in \mathbb{Z}_q^{2n \times 2n}\}$.
- 3) Choose four collision-resistant hash functions:

$$G_1 : \{0, 1\}^* \rightarrow \mathbb{Z}_q^n \quad (7)$$

$$G_2 : \{0, 1\}^N \rightarrow \mathbb{Z}_q^n \quad (8)$$

$$G_3 : \{0, 1\}^* \rightarrow \{0, 1\}^n \quad (9)$$

$$G_4 : \{0, 1\}^N \rightarrow \{0, 1\}^\tau \quad (10)$$

- 4) Set master secret key $MSK = B$ and use h as the master public key.
- 5) Output public parameters $params = (g, G_1, G_2, G_3, G_4)$ and keep system master private key MSK secret.

C. KeyGen($params, ID_i, MSK$)

Given public parameters $params$, user identity ID_i and master secret key MSK . The PKG computes a key pair (pk_i, sk_i) for the i -th user as follows:

- 1) Compute public key $t_i = G_1(ID_i) \in \mathbb{Z}_q^n$, set $pk_i = t_i$.
- 2) Randomly select $n_i \in R_q$, use CGS sampling algorithm to generate $(s_1, s_2) = (n_i, 0) - CGS(MSK, \sigma, (n_i, 0))$, such that $s_1 + s_2 g = n_i$.
- 3) Randomly choose polynomial vectors $s'_1, s'_2 \leftarrow D_n^\sigma$, and return user's private key $sk_i = (s_1, s_2, s'_1, s'_2)$.

Denote (pk_x, sk_x) and (pk_r, sk_r) as the public/private key pairs of signer and receiver respectively.

D. RingSigncryption($params, U, m, Sk_x, Pk_x$)

Given public parameters $params$, user identity set $U = (ID_1, ID_2, \dots, ID_N)$ and system master key MSK , where $ID_x \in U$, message $m \in \{0, 1\}^\tau$, user ID_x 's private key

$sk_x = (s_1, s_2, s'_1, s'_2)$ and public key pk_x . The ring signcryption by ID_x proceeds as follows:

- 1) Compute $J = n_x + n'_x + G_2(event) \in R_q$ as linkability tag, where $n_x = s_1 + s_2g \in R_q$, $n'_x = s'_1 + s'_2g \in R_q$. Randomly select event identifier $event \leftarrow \{0, 1\}^N$.
- 2) Choose short polynomial vectors $g_i, g'_i \leftarrow D_n^\sigma$, for $i \in \{1, 2, \dots, N\}$.
- 3) Compute $v = G_3\left(\sum_{i=1}^N h_i + h'_i h, U, m, J\right)$.
- 4) For $i = 1, 2, \dots, N$: if $i \neq x$, set $z_i = h_i, z'_i = h'_i$; if $i = x$, then

$$z^{(x)} = \begin{pmatrix} z_x \\ z'_x \end{pmatrix} = \begin{pmatrix} s_1 + s'_1 \\ s_2 + s'_2 \end{pmatrix} v + \begin{pmatrix} h_x \\ h'_x \end{pmatrix}. \quad (11)$$

- 5) To reduce correlation between $z^{(x)}$ and sk_x , $(\{z^{(i)}\}, h)$ is obtained via rejection sampling with probability

$$\frac{D_{\tilde{O}(2n\sigma)}(z_x, z'_x) / MD(((s_1 + s'_1)v, (s_2 + s'_2)v), D_{\tilde{O}(2n\sigma)}((z_x, z'_x)))}{M}, \quad (12)$$

where $M = O(1)$. Output signature $\text{Sig}(m) = ((z_i, z'_i)_{1 \leq i \leq N}, v, J)$. Otherwise restart.

- 6) Compute $c = m \oplus G_4(k \parallel event)$, where $k \leftarrow \{0, 1\}^N$.
- 7) Compute $u = rg + e_1 \in R_q$, $w = rpk_r + e_2 + q/2 \cdot k \in R_q$, with $r, e_1, e_2 \leftarrow [-1, 0, 1]^N$. Then truncate least significant bits of w : $w = 2^l \lfloor w/2^l \rfloor$, with $l \leq \log_2 q - 3$.
- 8) Output ciphertext $C = (v, w, c, \text{Sig}(m))$.

E. Unsigncryption(params, V, m, s₂, C, Event)

Given public parameters *params*, user identity set $U = (ID_1, ID_2, \dots, ID_N)$, message $m \in \{0, 1\}^\tau$, part of receiver's private key s_2 , ciphertext $C = (v, w, c, \text{Sig}(m))$, and event identifier *event*. Receiver ID_r does:

- 1) Compute $f = w - vs_2$, $k = fq/2$, $p = G_2(event)v$.
- 2) Compute $m = c \oplus G_4(k \parallel event)$.
- 3) Compute verification parameter $v' = G_3\left(\sum_{i=1}^N z_i + z'_i g + p - Jv, V, m, J\right)$. If $v' = v$, then $\text{Sig}(m)$ is valid.
- 4) If $\|z_i\| \leq 2\sigma n$, $\|z'_i\| \leq 2\sigma n$ and condition (3) holds, then ID_r outputs m ; otherwise output $\perp$.

5) Link(Sig(m₁), Sig(m₂))

Input two signatures $\text{Sig}(m_1)$ and $\text{Sig}(m_2)$. The verifier performs:

- 1) Verify whether $\text{Sig}(m_1)$ and $\text{Sig}(m_2)$ are valid signatures.
- 2) Check whether $J_1 = J_2$. If both conditions hold, output "Linked"; otherwise output "Unlink".

VI. SECURITY PROOF

A. Correctness Analysis

- 1) *Signature Correctness*: Suppose the signature

$$\text{Sig}(m) = ((z_i, z'_i)_{1 \leq i \leq N}, u, J) \quad (13)$$

is valid and signed by a member of the ring

$$V = (ID_1, ID_2, \dots, ID_N), \quad (14)$$

then the following equality holds:

$$\begin{aligned} & \sum_{i=1}^N (z_i + z'_i h) + p - Ju \\ &= \sum_{i \in \{1, 2, \dots, N\}, i \neq x} (z_i + z'_i h) + (z_x + z'_x g) + p - Jv \\ &= \sum_{i \in \{1, 2, \dots, N\}, i \neq x} (h_i + h'_i g) + (s_1 + s'_1)v + g_x \\ &= +[(s_2 + s'_2)v + g'_x]g + p - Ju \\ &= \sum_{i=1}^N (h_i + h'_i g) + (s_1 + s'_1)u + (s_2 + s'_2)ug + p - Jv \\ &= \sum_{i=1}^N (h_i + h'_i g) + (s_1 + s_2g)v + (s'_1 + s'_2g)u + p - Jv \\ &= \sum_{i=1}^N (h_i + h'_i g). \end{aligned} \quad (15)$$

Since $p = G_2(event)v$, $J = n_x + n'_x + G_2(event)$, it follows that

$$\sum_{i=1}^N (z_i + z'_i g) + p - Jv = \sum_{i=1}^N (h_i + h'_i g). \quad (16)$$

Thus,

$$\begin{aligned} & H_3\left(\sum_{i=1}^N (z_i + z'_i g) + p - Ju, V, m, J\right) \\ &= H_3\left(\sum_{i=1}^N (h_i + h'_i g), V, m, J\right), \end{aligned} \quad (17)$$

i.e., $u' = u$.

2) Unsigncryption Verification Process:

$$\begin{aligned} f &= w - us_2 \\ &= (rpk_s + e_2) - (rg + e_1)s_2 \\ &= r(s_1 + s_2g) + e_2 - (rg + e_1)s_2 \\ &= rs_1 + e_2 - e_1s_2. \end{aligned}$$

For correct decryption, all coefficients of $f = rs_1 + e_2 - e_1s_2$ must lie within $(-q/4, q/4)$. Thus, q must be sufficiently large. Note that bit loss may cause decryption errors. When $l \leq \log_2 q - 3$, it has no significant effect on correctness. Therefore, this value of l is taken as the standard.

3) *Linkability Verification Process*: Suppose the signer computes linkability tags J_1 and J_2 for messages m_1 and m_2 respectively. In the proposed scheme,

$$\begin{aligned} J_1 &= (s_1 + s_2g) + (s'_1 + s'_2g) + G_2(event), \\ J_2 &= (s_1 + s_2g) + (s'_1 + s'_2g) + G_2(event). \end{aligned} \quad (18)$$

Both are generated from the signer's public and private keys. Since J_1 and J_2 are computed using the same keys and event identifier (including the system master key), it follows that $J_1 = J_2$, satisfying linkability correctness.

VII. SECURITY ANALYSIS

This section proves that NTRU-GIBLRSCS satisfies confidentiality, unforgeability, unconditional anonymity, and linkability.

Theorem 2. If adversary δ cannot break the confidentiality guarantee of NTRU-GIBLRSCS in the ROM, then it is assumed that the proposed scheme satisfies confidentiality. In the ANON-sID-CPA game, suppose δ wins with advantage ε . Then the challenger γ can solve the DRLWE problem with advantage $\varepsilon' \geq \varepsilon/eq^k$, where e denotes the base of the natural logarithm, and q^k denotes the number of queries to the Corruption Oracle (CO).

Proof. Assume γ is the adversary receiving an instance of the DRLWE problem, whose goal is to determine whether $(a, b = ax + e)$ is from distribution $A_{x,\chi}$ or uniformly distributed over $R_q \times R_q$. The challenger γ interacts with the subroutine δ in the ANON-sID-CPA game as follows:

- 1) **Setup phase:** γ runs system setup and key extraction algorithms to obtain public parameters $params$, and returns them along with pk_i as the public key of user ID_i to δ . Then, δ performs adaptive queries.
- 2) **Hash queries:**
 - a) H_1 query: δ chooses user $ID_i \in R$, and γ returns pk_i .
 - b) H_2 query: γ randomly chooses $event \leftarrow \{0, 1\}^N$, and returns linkability tag J .
 - c) H_3 query: δ inputs message $m \in \{0, 1\}^\tau$ and identity set $U = (ID_1, ID_2, \dots, ID_N)$, then randomly chooses short polynomial vectors $g_i, g'_i \leftarrow D_n^\sigma$, $i \in \{1, 2, \dots, N\}$. γ randomly selects an integer v and returns it.
 - d) H_4 query: γ randomly selects $k \leftarrow \{0, 1\}^N$, and returns ciphertext C .
- 3) **Private key query:** δ queries the key of ID_i . If $ID_i \neq ID_\alpha$, then γ stops, where ID_α is the target identity of the challenge; otherwise, γ returns sk_i .
- 4) **Ring signcryption query:** δ requests signcryption on m . γ responds:
 - a) Compute $J = pk_1 + pk_2 + G_2(event) \in R_q$, randomly choose $h_i, h'_i \leftarrow D_n^\sigma$.
 - b) Compute $u = G_3\left(\sum_{i=1}^N h_i + h'_i g, V, m, J\right)$.
 - c) For $i = 1, 2, \dots, N$: if $i \neq x$, set $z_i = g_i, z'_i = h'_i$; if $i = x$, then

$$z^{(x)} = \begin{pmatrix} z_x \\ z'_x \end{pmatrix} = \begin{pmatrix} s_1 + s'_1 \\ s_2 + s'_2 \end{pmatrix} v + \begin{pmatrix} g_x \\ g'_x \end{pmatrix}. \quad (19)$$

- d) With minimal probability

$$\frac{D_{\tilde{O}(2n\sigma)}(z_x, z'_x)}{MD(((s_1 + s'_1)v, (s_2 + s'_2)v), D_{\tilde{O}(2n\sigma)}((z_x, z'_x)))}, \quad M = O(1), \quad (20)$$

output $\text{Sig}(m) = ((z_i, z'_i)_{1 \leq i \leq N}, u, J)$.

- e) Compute $c = m \oplus G_4(k \parallel event)$, where $k \leftarrow \{0, 1\}^N$.
- f) Compute $v = rg + e_1 \in R_q$, $w = rpk_r + e_2 + q/2 \cdot k \in R_q$, with $r, e_1, e_2 \leftarrow \{-1, 0, 1\}^N$. Then truncate: $w = 2^l \lfloor w/2^l \rfloor$, with $l \leq \log_2 q - 3$.

g) Output ciphertext $C = (v, w, c, \text{Sig}(m))$.

5) Unsigncryption query:

- a) Compute $f = w - vs_2, k = fq/2$.
- b) Compute $m = c \oplus G_4(k \parallel event)$.
- c) Compute $u' = G_3\left(\sum_{i=1}^N z_i + z'_i g + t - Ju, V, m, J\right)$. If $u' = u$, then $\text{Sig}(m)$ is valid.
- d) If $\|z_i\| \leq 2\sigma n$, $\|z'_i\| \leq 2\sigma n$ and condition (3) holds, then ID_r outputs m ; otherwise $\perp$.

6) **Challenge query:** δ issues challenge query on (m, ID_0^*, ID_1^*) . If δ has never queried KeyGen for ID_0^*, ID_1^* , then γ continues; otherwise aborts. γ randomly chooses $\theta \in \{0, 1\}$ and computes challenge ciphertext $C^* = (u^*, w^*, c^*, \text{Sig}^*(m^*))$ for ID_θ^* .

7) **Guessing phase:** δ continues adaptive queries, and finally outputs θ^* as guess.

Analysis: γ knows (u^*, h^*) and (w^*, pk_r^*) come from distribution $A_{x,\chi}$. If $\theta^* = \theta$ and $e_1^*, e_2^* \leftarrow \{-1, 0, 1\}^N$, then $A_{x,\chi} = ((v^*, g^*), (w^*, pk_r^*))$ is a solution to DRLWE. Let φ be probability $ID_i = ID_\alpha$. The probability γ does not fail in both phases is φq^k . In challenge phase, probability is $1 - \varphi$. Thus, success probability is $\varepsilon \varphi q^k (1 - \varphi)$. When $\varphi = 1 - 1/(1 + q^k)$, success probability is at least $\varepsilon/(eq^k)$. This is negligible for γ to solve DRLWE.

VIII. UNCONDITIONAL ANONYMITY

Theorem 3. Assuming NTRU-GSIS is unforgeable, this scheme guarantees unforgeability.

Proof: Suppose ϕ successfully forges a signature with a non-negligible probability ϵ . It can be shown that γ , using the output of ϕ 's forgery, can determine a set of non-zero small polynomials $(v, u) \in R_q^2$ satisfying

$$v + u * g = 0 \pmod{q}, \quad (21)$$

thus constructing a solution to the SIS problem on the post quantum Lattice cryptography. To keep the exposition concise, it is not elaborated here; interested readers may refer to [30] for the full proof.

A. Unconditional Anonymity

Theorem 4. This scheme is unconditionally anonymous.

Proof: Unconditional anonymity is proven via a game between the attacker δ and the challenger γ .

- 1) **Setup phase:** This phase is the same as the first phase in Theorem 2.
- 2) **Adaptive queries:** These queries are the same as those in the second phase of Theorem 2.
- 3) **Challenge phase:** δ inputs a message $m^* \in \{0, 1\}^\tau$ and a set of user identities

$$V^* = (ID_1^*, ID_2^*, \dots, ID_N^*), \quad (22)$$

as well as two user identities $ID_i^*, ID_j^* \in R^*$. γ chooses a random bit $b \in \{0, 1\}$, computes the private key sk_i^* corresponding to ID_b^* , runs the RingSigncryption algorithm, and returns

$$C^* = (v, w, c, \text{Sig}^*(m^*)) \quad (23)$$

as the ciphertext.

- 4) **Guess phase:** δ guesses $b^* \in \{0, 1\}$ under the condition that ID_i^* and $ID_j^* \in R^*$ have not been queried by CO or SO so far.

Analysis: The advantage of δ winning is

$$\text{Adv}_\delta^{\text{anon}} = \left| \Pr[b^* = b] - \frac{1}{2} \right| = \epsilon, \quad (24)$$

which is negligible.

The distribution of $C^* = (v, w, c, \text{Sig}^*(m^*))$ generated by the user ID_i^* 's sk_i is computationally indistinguishable from

$$C' = (v, w, c, \text{Sig}'(m')) \quad (25)$$

generated by the user ID_j^* 's sk_j of the challenger γ .

According to Lemma 2 and Theorem 1,

$$z^*(b) = (z_i^*, (z_i')^*) \quad (26)$$

is statistically close to $(D_n^\sigma)^2$ (specifically, the statistical distance is randomly distributed as $2^{-\omega(\log n)/M}$).

Therefore, $C^* = (v, w, c, \text{Sig}^*(m^*))$ and $(D_n^\sigma)^{2N}$ are indistinguishable. Similarly, $C' = (v, w, c, \text{Sig}'(m'))$ and $(D_n^\sigma)^{2N}$ are also indistinguishable. The indistinguishability between C^* and C' is evident.

Hence, NTRU-GIBLRSCS satisfies the unconditional anonymity requirement of ring signcryption.

B. Linkability

Theorem 5. The proposed NTRU-GIBLRSCS is linkable.

Proof: Based on the following assumption: δ wins the linkability game (see Section 3.2.4 and Definition 7), and ϵ is a non-negligible probability.

- 1) **Setup phase:** This phase is the same as the first phase in Theorem 2.
- 2) **Adaptive queries:** These queries are the same as those in the second phase of Theorem 2. δ outputs two signatures satisfying the following conditions:
 - a) All public key elements V_1^*, V_2^* are generated via the RO oracle.
 - b) $\text{Sig}^*(m_1^*), \text{Sig}^*(m_2^*)$ are not obtained via SO queries.
 - c) CO is queried at most once.

Analysis: Suppose using a single private key, δ can generate two valid signatures $\text{Sig}(m_1)$ and $\text{Sig}(m_2)$ with non-negligible probability. Since the scheme is unforgeable, the algorithm can only verify these two signatures if δ honestly uses its private key to generate $\text{Sig}(m_1)$ and $\text{Sig}(m_2)$.

In simple terms,

$$\begin{aligned} J_1 &= nx + nx' + G_2(\text{event}) \\ &= (s_1 + s_2 * g) + s'_1 + s'_2 * g + G_2(\text{event}) \end{aligned} \quad (27)$$

and

$$\begin{aligned} J_2 &= \tilde{n}x + \tilde{n}x' + G_2(\text{event}') \\ &= (s_3 + s_4 * g) + (s'_3 + s'_4 * g) + G_2(\text{event}'). \end{aligned} \quad (28)$$

Moreover, since a single public key corresponds to this signing key, i.e.,

$$(s_1, s_2, s'_1, s'_2) = (s_3, s_4, s'_3, s'_4), \quad \text{and} \quad \text{event} = \text{event}', \quad (29)$$

TABLE II
NOTATIONS FOR SIGNCRYPTION OPERATIONS

Symbol	Signcryption Operation
T_{SD}	Discrete Gaussian Sampling (DGS) Algorithm
T_{RS}	Rejection Sampling Algorithm
T_1	Polynomial $\times$ Polynomial Operation
T_2	Matrix $\times$ Vector Operation
T_H	Hash Computation
T_I	Image Sampling Algorithm

and the polynomial $h \in R_q$ is a randomly chosen public parameter, we have

$$J_1 = J_2. \quad (30)$$

This shows that the algorithm

$$\text{Link}(\text{Sig}(m_1), \text{Sig}(m_2)) \quad (31)$$

returns “Linked”. Therefore, the advantage of δ , $\text{Adv}_\delta^{\text{link}}$, is negligible.

IX. PERFORMANCE ANALYSIS

This section evaluates the efficiency of the proposed NTRU-GIBLRSCS scheme by comparing it with existing lattice-based ring signcryption schemes [25], [29] and NTRU-lattice-based ring signature schemes [4], [28], focusing primarily on *time cost* and *storage overhead*. In order to generate user signatures with robust anonymity, the technique integrates rejection sampling, uses compact Gaussian sampling to obtain private keys for ring members, and uses a trapdoor generation process to establish the master keys. Additionally, identity protection during encryption is guaranteed by a key encapsulation approach. The Random Oracle Model (ROM) security study shows the scheme's linkability property and resilience to forgery.

Table II summarizes the notation used for encryption operations, while Table III presents the time costs for signature generation, verification, encryption, and decryption, along with identity-based characteristics. In our approach, each member contributes a public verification component derived from the underlying post-quantum lattice IBE parameters, and the ring is algebraically created by aggregating the public identity-derived keys of all possible signers into a cyclic structure. These components are incorporated by the signcryption method into a series of algebraic relations that must all be satisfied at the same time for the ciphertext to be legitimate. In order to ensure that the system of equations remains satisfiable regardless of which index belongs to the true signer, one of these relations is instantiated for the actual signer using the personal secret key, while the rest relations are filled with simulator-generated values.

In this analysis, operations with negligible runtime—such as hash evaluations, matrix- G addition, and polynomial- G addition—are omitted. The comparison instead emphasizes the dominant cost factors, namely matrix- G vector multiplications and polynomial- G polynomial multiplications.

For the proposed scheme, signature generation involves the double rejection sampling algorithm, $\alpha + 1$ polynomial

TABLE III
ANALYSIS OF TIME COSTS AND IDENTITY-BASED TRAIT

Scheme	Signature Generation	Verification	Encryption	Decryption	Identity-Based
Cao et al. [25]	$\alpha T_{SD} + (\alpha + 1)T_2 + T_I$	αT_2	$3T_2 + 4T_{SD}$	T_2	No
Saeed et al. [29]	$\alpha T_{SD} + (2\alpha + 1)T_2 + T_I$	$(\alpha + 1)T_2$	$2T_2 + 3T_{SD}$	T_2	No
Castiglione et al. [4]	$2(\alpha + 1)T_{SD} + 2(\alpha + 1)T_1$	$2\alpha T_1$	—	—	No
Hafeez et al. [28]	$2\alpha T_{SD} + (\alpha + 1)T_1 + 2T_{RS}$	αT_1	—	—	Yes
Proposed Scheme	$2\alpha T_{SD} + (\alpha + 1)T_1 + 2T_{RS}$	αT_1	$2T_1$	T_1	Yes

multiplications, and 2α Gaussian sampling operations. Thus, the cost of signature generation is:

$$2\alpha \cdot T_{SD} + (\alpha + 1) \cdot T_1 + 2 \cdot T_{RS}. \quad (32)$$

Verification requires only α polynomial multiplications, giving:

$$\alpha \cdot T_1. \quad (33)$$

Compared to Castiglione et al. [4], the proposed scheme achieves higher efficiency in both signature generation and verification. The intrinsic characteristics of ring signatures, which guarantee that the verifier can verify a legitimate signature from one member of the ring without identifying which one, preserve anonymity in ring signcryption methods throughout many signcryptions from the same user. Because each signature appears computationally indistinguishable from those generated by any ring member, the mathematical structure prevents linking specific signcryptions to a single identity even when an adversary observes multiple signcryptions shared by the same ring or produced by the same signer (linked by message patterns or timing). This is true under adaptive chosen-message attacks, where the signer's position is protected by new randomness each signcryption. Moreover, since it is identity-based, it eliminates the overhead of certificate management. This significantly improves scalability, as large-scale IoT deployments with many ring members would otherwise face resource bottlenecks due to certificate handling. Traceability and linkability are two different ideas, especially when it comes to data management and privacy-preserving technologies. Conventional PKI-based cryptosystems rely on a complicated certificate lifecycle that includes issuing, distributing, checking, updating, and revoking digital certificates, they add significant overhead to large-scale IoT installations. Memory requirements are increased since every IoT device needs to keep both its own certificate and frequently the certificates of reliable authority. Additionally, many limited devices cannot afford the constant connectivity and processing resources required for certificate validation, particularly when using online protocols like OCSP or CRLs. This certificate management load scales poorly as the number of devices increases, leading to increasing administrative complexity, greater latency, and higher communication costs.

Although the computational cost of signature generation and verification matches that of Hafeez et al. [28], our scheme integrates encryption into the process. This provides signcryption functionality without increasing signature size—an advantage absent in Castiglione et al. [4].

For encryption and decryption costs, the proposed scheme requires two and one polynomial- G multiplications, respectively:

$$2 \cdot T_1 \quad \text{and} \quad T_1. \quad (34)$$

In contrast, Cao et al. [25] incurs:

$$3 \cdot T_2 + 4 \cdot T_{SD} \quad \text{and} \quad T_2, \quad (35)$$

while Saeed et al. [29] requires:

$$2 \cdot T_2 + 3 \cdot T_{SD} \quad \text{and} \quad T_2. \quad (36)$$

Since $T_1 < T_2$, it follows that:

$$2 \cdot T_1 < 2 \cdot T_2 < 3 \cdot T_2. \quad (37)$$

Thus, the proposed scheme offers significant efficiency improvements over [25], [29].

A. Storage Overhead

Table IV compares the storage requirements across schemes, considering public/private keys and signature/ciphertext lengths. In our scheme, the public key is a β -dimensional vector, while the private key comprises four small polynomials in the ring R_γ . Thus:

$$\text{Public Key Size} = \beta \log \gamma, \quad (38)$$

$$\text{Private Key Size} = 4\beta \log \gamma. \quad (39)$$

From Table IV, it is evident that the public key length in the proposed scheme is shorter than in [25] and [29], and while equal to [4] and [28], those schemes lack integrated encryption functionality.

Although the private key is longer than in [25] and [29], our design reduces encryption and decryption overhead. The output signature of the proposed scheme is:

$$\text{Sig}(m) = ((z_i, z'_i)_{1 \leq i \leq \alpha}, u, J), \quad (40)$$

where $(z_i, z'_i, J) \in R_\gamma$. Thus, the signature length is:

$$(2\alpha + 1) \log \gamma, \quad (41)$$

which equals [4] but is smaller than [28].

Although ciphertext and signature lengths exceed those of [25] and [29], the proposed scheme achieves shorter public keys, which is advantageous for lightweight IoT deployments.

TABLE IV
COMPARISON OF STORAGE OVERHEAD

Scheme	Public Key	Private Key	Signature	Ciphertext
Cao et al. [25]	$(\beta \times k) \log \gamma$	$(m \times k) \log \gamma$	$\alpha m \log \gamma + k \log \gamma$	$\alpha m \log \gamma + 2\beta \log \gamma + k \log \gamma$
Saeed et al. [29]	$(\beta \times m) \log \gamma$	$(m \times k) \log \gamma$	$\alpha \beta \log \gamma + 3m \log \gamma$	$(\alpha + 2)\beta \log \gamma + 3m \log \gamma$
Hafeez et al. [28]	$\beta \log \gamma$	$9\beta \log \gamma$	$(3\alpha + 1)\beta \log \gamma$	—
Castiglione et al. [4]	$\beta \log \gamma$	$4\beta \log \gamma$	$(2\alpha + 1)\beta \log \gamma$	—
Proposed Scheme	$\beta \log \gamma$	$4\beta \log \gamma$	$(2\alpha + 1)\beta \log \gamma$	$(2\alpha + 3)\beta \log \gamma$

TABLE V
PARAMETER SETTING

Type	n	m	k	q
Type 1	256	288	3804	1790
Type 2	256	288	3804	1712

TABLE VI
COMPARISON OF TIME COSTS (MS)

Parameter	Cao et al.[25]	Saeed et al. [29]	Proposed Scheme
Encryption under $\lambda = 90$	328.702	220.403	33.0
Decryption under $\lambda = 90$	104.501	104.501	16.5
Encryption under $\lambda = 128$	570.343	381.083	33.0
Decryption under $\lambda = 128$	186.701	186.701	16.5

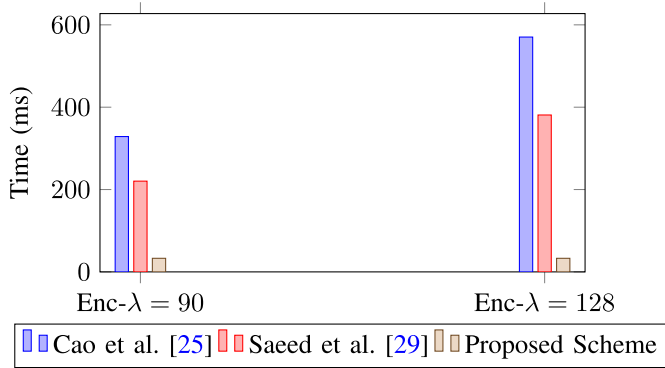


Fig. 2. Comparison of encryption time (ms) between existing schemes and the proposed scheme under $\lambda = 90$ and $\lambda = 128$. The proposed scheme demonstrates over 90% reduction in encryption time.

X. RESULT DISCUSSION

Table V lists the parameter settings of the proposed scheme. The simulation experiment was conducted on a Windows 11 operating system, with an AMD Ryzen 5 4500U processor and 16 GB of memory. The experimental data are the averages of 1000 operations.

Table VI lists the encryption and decryption times of the proposed scheme and schemes [25], [29] under Type 1 and Type 2 parameter selections, mainly based on the parameters and execution times referenced in [4]. The values in the table confirm the analysis based on Table III.

Figures 2 and 3 present a visual comparison of the encryption and decryption times for the proposed NTRU-GIBLRSCS scheme against existing works [25], [29]. The bar chart in Figure 2 clearly shows that, for both $\lambda = 90$ and $\lambda = 128$, the proposed scheme significantly outperforms Cao et al. [25]

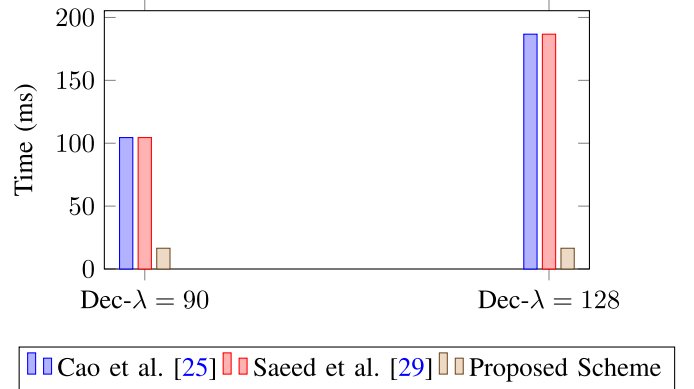


Fig. 3. Comparison of decryption time (ms) between existing schemes and the proposed scheme under $\lambda = 90$ and $\lambda = 128$. The proposed scheme achieves an average reduction of 87.6%.

and Saeed et al. [29]. Specifically, encryption costs are reduced by more than 90% on average, demonstrating that the proposed construction is computationally lightweight and scalable. Through putting processing closer to the data source, edge computing plays a critical role in enhancing computational efficiency and data security inside IoT networks. First, edge computing greatly minimises the exposure of sensitive data to external networks by carrying out processing and preliminary analysis at or near the device level. Data can be locally filtered, encrypted, or anonymised rather than sent to centralised cloud servers in its original form. This reduces the likelihood of interception, reduces attack surfaces, and speeds up the identification of questionable activity. Before data leaves the device environment, edge nodes can also handle local authentication, enforce security policies, and perform simple cryptographic operations.

Similarly, Figure 3 illustrates the decryption time comparison. While the baseline schemes require over 100 ms for decryption under both security levels, the proposed scheme achieves consistent performance at 16.5 ms, leading to an average reduction of 87.6%. The lattice dimension (n) and noise variance (or magnitude) have a direct trade-off with the total security level and execution time in lattice-based cryptography, and they have a strong, inverse relationship with one another. For effective polynomial multiplication the primary operation in many ideal lattice schemes use optimization techniques such as the Number Theoretic Transform. Investigate methods that substitute deterministic mistakes for random noise to streamline computation at the possible expense of certain security presumptions. The suggested design specifically supports

TABLE VII
COMPARISON OF STORAGE OVERHEAD

Scheme	Cao et al.[25]	Saeed et al. [29]	Hafeez et al. [28]	Castiglione et al. [4]	Proposed Scheme
Public Key Length	1.031	1.109	111	–	–
Private Key Length	0.140	0.140	944	–	–
Signature Length ($N = 1$)	0.140	1.328	433	–	–
Signature Length ($N = 8$)	0.906	8.328	2517	1717	–
Signature Length ($N = 64$)	7.031	64.328	193129	129129	–
Signature Length ($N = 128$)	14.031	128.328	385257	257	–
Signature Length ($N = 264$)	28.031	256.328	769513	513	–
Ciphertext Length ($N = 1$)	2.140	3.328	–	–	5
Ciphertext Length ($N = 8$)	2.906	10.328	–	–	19
Ciphertext Length ($N = 64$)	9.031	66.328	–	–	131
Ciphertext Length ($N = 128$)	16.031	130.328	–	–	259
Ciphertext Length ($N = 264$)	30.031	258.328	–	–	515

linkability without permitting complete traceability, ensuring signer anonymity while maintaining structural responsibility. Anybody (or a designated verifier) can ascertain whether two ciphertexts or signatures were produced by the same, anonymous signer thanks to linkability. Key Property: Linkability reveals the relationship between outputs without disclosing the signer's name or ring position. The goal is to protect anonymity while preventing fraud, double-signing, double-encryption, and misuse.

The drastic reduction in both encryption and decryption time confirms that the proposed NTRU-GIBLRSCS scheme is well-suited for *resource-constrained IoT consumer electronics devices*, where low latency and predictable performance are essential. Ciphertext indistinguishability under chosen-ciphertext attacks (IND-CCA) ensures strong security by preventing information leakage even when attackers actively probe the system's decryption. This means that an adversary cannot determine whether a ciphertext encrypts one chosen message or another (or random noise) better than guessing, even with a decryption oracle and the ability to query ciphertexts. This is accomplished by employing strategies like padding (such as OAEP), randomness, and hash functions, which make the ciphertext appear random and guard against attacks that take advantage of particular message patterns or decoding problems. Significant part to the intrinsic limits of IoT devices and the increasing complexity of contemporary security needs, secure IoT communication frameworks face a number of interrelated issues in balancing computing overhead, energy consumption, and latency. First, there is frequently a significant computational overhead associated with putting strong cryptographic techniques into practice, particularly those intended for post-quantum security. IoT devices often have limited memory, constrained processors, and little hardware acceleration, which makes it challenging to support key-exchange protocols, heavyweight encryption, or signature generation without compromising system performance. Second, increased computing directly affects energy usage, which is important for battery-operated gadgets like environmental monitors, wearables, and sensors. Large key sizes and frequent sample steps in cryptographic operations quickly deplete energy, shortening device life and raising maintenance expenses. These results not only validate the theoretical efficiency analysis but also demonstrate practical

feasibility for real-world deployment in applications such as secure communication in smart home devices, wearable sensors, and consumer-grade embedded systems.

Compared with scheme [25], the encryption time is reduced by an average of 92.087%, and the decryption time is reduced by an average of 87.687%. Compared with scheme [29], the encryption time is reduced by an average of 88.184%, and the decryption time is reduced by an average of 87.687%. Overall, the encryption time is reduced by an average of 90.136%, and the decryption time is reduced by an average of 87.687%.

Table VII lists the public key/private key lengths and signature/ciphertext lengths of the proposed scheme and schemes [4], [25], [28], [29] under Type 1 parameter selection. Inspired by references [21], [22], the value of m in the proposed scheme is set to 28, because $m > 5n \log q$ and $m < 30$.

The values in the table confirm the analysis based on Table IV. Through intricate constructions that connect the ciphertext's structure to the plaintext often involving randomisation, specialised encoding, and semantic security non-malleability in cyphers prevents attackers from altering the value of m in the proposed scheme is set to a ciphertext to produce a meaningful, related plaintext without knowing the original key, guaranteeing that any modification results in an unrelated or unreadable message. When a substantial amount of random noise or padding is added to the plaintext prior to encryption, the ciphertext becomes heavily reliant on the random element rather than merely the message. Compared with scheme Cao et al. [25], the public key length is reduced by an average of 3.007%, and compared with scheme Saeed et al. [29], the public key length is reduced by an average of 9.829%. Overall, the public key length is reduced by an average of 6.418%.

XI. CONCLUSION

In this paper, rejection sampling and compact Gaussian sampling (CGS) techniques are combined. Based on the DRLWE and NTRU small integer solution (SIS) problems as underlying hard problems, an identity-based linkable ring signcryption scheme over the post quantum Lattice cryptography, NTRU-GIBLRSCS, is proposed. The study's limitations should be briefly explored and presented as potential areas for further investigation in the conclusion. In addition to demonstrating academic integrity, this method aids readers in comprehending

the precise circumstances and extent of the results' validity. We suggest an identity-based linkable ring signcryption approach built on post quantum Lattice cryptography specifically for edge-enabled Internet of Things devices in order to overcome these difficulties. The security of the NTRU-GIBLRSCS scheme is analyzed in detail under the random oracle model, proving that the system satisfies correctness, unforgeability, unconditional anonymity, and linkability requirements. Performance analysis and experiments show that, among existing lattice-based signcryption schemes, the proposed scheme has smaller public key length and higher encryption/decryption efficiency. Furthermore, in NTRU-lattice-based ring signatures, the scheme has smaller key and signature lengths. Future research directions include exploring methods to further reduce ciphertext length.

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